Project Design Phase-I
Solution Architecture

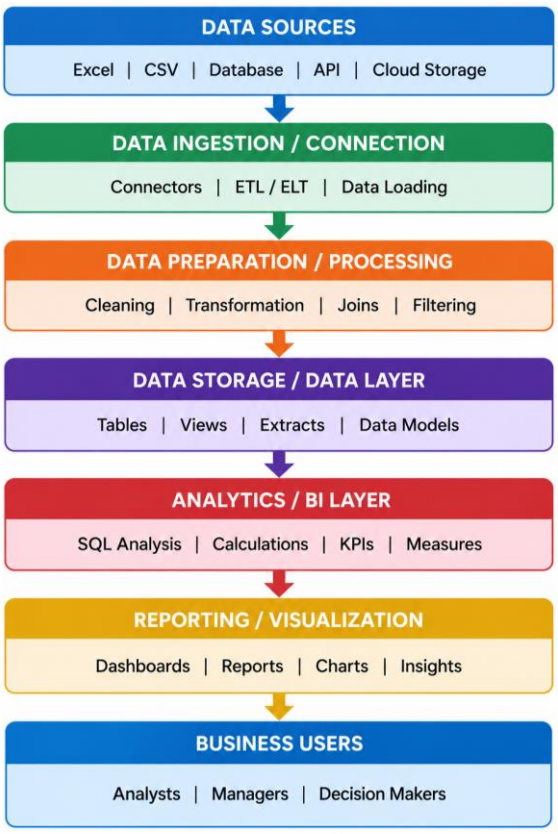
Date	21 October 2026
Team ID	EN25cs3010849
Project Name	Global Olympic
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Sample Solution Architecture Diagram:



Component Description Table

Component Name	Description/Role in Architecture	Technology Used
Data Sources	Stores the foundational historical Olympic datasets containing medal tallies, athlete profiles, and event details.	CSV Files (medals.csv, athlete.csv, athlete_events.csv)
Data Ingestion / Connection	Connects raw files and loads records into the workspace for processing and pipeline execution.	Python File Connectors / Pandas Ingestion
Data Preparation / Processing	Cleans raw data, handles missing values, removes duplicates, and merges datasets based on common keys.	Python (Pandas, NumPy) & SQL